# When Mutualism Becomes Costly: Abiotic Stress Shapes Grass Demography

Jacob K. Moutouama *[1,2,3], Julia Martin[1], Dajanae Pearson[1], Alexandra Jimenez Martín[1], Josh Fowler[1,4], Ali Campbell[1], Chris Oxley[1], Karl Schrader[1], and Tom E.X. Miller[1]

[1]Program in Ecology and Evolutionary Biology, Department of BioSciences, Rice University, Houston, TX USA
[2]Department of Botany, University of British Columbia, Vancouver, BC, Canada
[3]Biodiversity Research Centre, University of British Columbia, Vancouver, BC, Canada
[4]University of Miami, Department of Biology, Miami, FL USA

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*Corresponding author: jmoutouama@gmail.com

Abstract

Beneficial interactions between plants and fungi are ubiquitous and foundational to terrestrial ecosystems. Plant-fungal mutualisms are expected to strengthen under environmental stress, yet the relative importance of abiotic and biotic factors in eliciting mutualistic benefits remains unclear. We studied how the demographic effects of seed-transmitted fungal endophytes on native three grass host species varied in response to abiotic and biotic factors using geographically distributed field experiments along an aridity gradient, cross with herbivore access or exclusion. Vital rate models showed that all components of host fitness (survival, growth and fertility) declined with increasing, indicating reduced performance under abiotic stress. Declines were more pronounced in individuals hosting endophytes, suggesting that interactions that are beneficial under benign conditions can become costly under abiotic stress. On the other hand, herbivory had no detectable effect on host demography or on the demographic effects of hosting endophytes. Our results indicate that abiotic stress was the primary determinant of interaction outcomes between plants and fungal symbionts, but in the opposite direction as predicted by theory and prior work. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

# Introduction

Plant-microbe symbioses are widespread and ecologically important. These interactions are famously context-dependent, where the direction and strength of the interaction outcome depend on the environment in which it occurs (Bronstein, 1994; Hoeksema and Bruna, 2015). For example, plant-microbe interactions that are beneficial under stressful conditions may become neutral or parasitic under benign conditions (Giauque et al., 2019).

Stress may derive from biotic and/or abiotic factors. Under biotic stress, such as herbivory, microbial symbiosis can benefit host plants by facilitating the production of secondary compounds that deter feeding or exert direct toxicity, thereby reducing herbivore growth, survival, and oviposition (Atala et al., 2022; Bastias et al., 2017; Vega, 2008). Similarly under abiotic stress, such as low precipitation or high temperature, symbionts can increase their host tolerance (Clay and Schardl, 2002)[1]. Mycorrhizal fungi form extensive networks that increase root surface area, helping plants absorb more water and nutrients such as phosphorus and nitrogen (Amijee et al., 1989). [2] Symbiotic microbes can also stimulate plants to produce antioxidant enzymes like superoxide dismutase and catalase that reduce oxidative damage caused by stresses such as drought, salinity, and heat (Ruiz-lazona et al., 1996). All these mechanisms could help the host species better buffer against environmental stochasticity (Fowler et al., 2023).

In resource-limited environments, maintaining symbionts can also impose significant costs on hosts (Bronstein, 2001; Johnson et al., 1997). [3] Because microbes depend on host-derived carbon and nutrients, plants must allocate part of their limited resources to sustaining the symbiont. This investment can reduce host growth, reproduction, or competitive ability, particularly when the benefits provided by the symbiont do not outweigh these costs (Klironomos, 2003).

Context-dependence[4] raises the hypothesis that plant–microbe interactions are likely to vary across abiotic stress gradients, such as high temperatures or low precipitation, as well as under biotic stressors like herbivory, with significant implications for host fitness in variable environments. If microbial symbiosis provides greater benefits under these stressful

[1] *Can you expand this a bit? The previous sentence gave a more thorough explanation of how symbionts can benefit hosts exposed to herbivory. How do symbionts promote tolerance of abiotic stress? Through this section, keep the scope pretty wide, so don't just focus on the grass-endophyte literature.*

[2] *Unclear how this relates to the preceding sentences. Is this meant to be an example of how symbionts promote tolerance of abiotic stress? If so, it does not come through clearly. What is the importance of access to phosphorous and nitrogen, and how does this related to abiotic stress?*

[3] *You are introducing a new idea, so I suggest starting a new paragraph with a strong topic sentence, and more fully developing the idea of costs of symbiosis.*

[4] *What is the topic of this paragraph, and what is it intended to do differently than the preceding paragraphs? A lot of this material overlaps heavily with what you have already presented above. Think about the logic structure that you want for the intro, and make sure your paragraphs are taking readers through a logical progression of ideas.*

conditions, symbionts could enhance host survival, growth, and reproduction.[5] (Allsup et al., 2023; Rudgers et al., 2020). For example, fungal endophytes of grasses enhance host tolerance to abiotic stress in *Bromus laevipes* by improving population survival in dry environments (Afkhami et al., 2014; David et al., 2019). Even when symbionts do not directly increase survival under stress, they may still promote host fitness by enhancing growth and reproduction, thereby offsetting the negative effects of lower survival rates (Yule et al., 2013). Conversely, if microbial symbiosis becomes costly under stressful conditions, symbionts could reduce host fitness[6] (Bennett and Groten, 2022; Benning and Moeller, 2021a,b). For instance, mycorrhizal fungi require substantial carbon subsidies from the host plant (up to 20% of photosynthetically fixed carbon) (Soudzilovskaia et al., 2015; Thirkell et al., 2020). When photosynthesis is limited by abiotic stress, this carbon cost may outweigh the benefits of symbiosis, leading to reduced host fitness.

Despite growing interest in the role of symbiosis in host performance, few studies have examined under natural conditions how both abiotic and biotic stressors shape the effects of symbionts on host demography along stress gradients (Louthan et al., 2015). Moreover, it remains unclear which stressor is most critical in eliciting the benefits of endophyte symbiosis[7], or how multiple stressors may interact to influence host demography. Herbivory is a biotic factor that can significantly affect host fitness (Benning et al., 2019) and mediate the outcomes of symbiotic interactions. Herbivores reduce host fitness through tissue damage, defoliation, and altered resource allocation (Maron and Crone, 2006), and can also impose indirect ecological costs by influencing reproductive or growth strategies (Miller et al., 2008). These ecological costs may shape the selective landscape in ways that favor symbiont persistence or transmission (Clay et al., 2005). For example, herbivory can enhance the fitness benefits of endophyte infection by triggering host defense pathways or promoting vertical transmission of the symbiont (Agrawal et al., 1999; Gundel et al., 2020). If the ecological context of herbivory alters the net fitness effects of symbiosis, then interactions between herbivory and abiotic stress (e.g., drought or high temperature) may play a critical role in determining host population persistence.[8]

[5] *Statements like this are a little circular ("if symbionts are beneficial then the plant would benefit.") Can you sharpen this?*

[6] *Also circular.*

[7] *What is the intended scope at this point in the paper? Is this meant to be a general statement about microbial symbioses, or are we now talking specifically about Epichloe symbioses in grasses? I think the Intro can provide clearer sign-posting for this. There is a lot of literature on drought and herbivory in the Epichloe literature. I think it would be good to have a paragraph late in the Intro that introduces this system and briefly covers the literature on drought and herbivory, highligting that few (or no?) previous studies have looked at both.*

[8] *The opening sentence suggested that this paragraph would be about the combined and relative roles of abiotic and biotic stress, but most of this paragraph is actually about herbivory. Again, bring attention to the way you structure paragraphs. Use clear topic sentences, and make sure the content of the paragraph stays true to the topic sentence.*

[9]

Working across an aridity gradient in the south-central US, we investigated how endophyte symbiosis influences host-plant demography under varying abiotic stress[10]. This region provides an ideal system to study biologically realistic stress gradients because it spans a strong east–west precipitation gradient, with mesic conditions in the east transitioning to increasingly arid conditions in the west. At each site along the gradient we also tested herbivory as a biotic stressor to evaluate its potential interactive effects with abiotic conditions. Specifically, we studied the symbiotic association between three native cool-season grass species (*Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*) and their vertically transmitted fungal symbionts (*Epichloë* spp.). Our experiment was designed to address the following questions[11]:

1. Under combined abiotic stress and herbivory, do endophytes provide fitness benefits or impose costs on their host plants?
2. Between abiotic and biotic factors, which plays a stronger role in shaping host population demography in association with endophytes[12]?

We hypothesized...[13]

# Materials and methods

## Study system

Our study focused on three cool-season perennial grasses native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a small, short-lived perennial that germinates from autumn to late winter and typically flowers between March and July. *E. virginicus* is a larger, long-lived species that begins flowering as early as March or April and continues throughout the summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers from late spring through summer. All three species host *Epichloë* fungal endophytes, which are primarily transmitted vertically through seeds. *Agrostis hyemalis* is associated with

[9] *Here is where I might dedicate a paragraph to describing the biology and context-dependence of the grass-endophyte symbioses.*

[10] *This paper includes multiple vital rate responses, and it might be strategic to highlight in the Intro the value of looking at multiple vital rates, since they may respond differently to abiotic or biotic stress (I am thinking of the demographic compensation ideas).*

[11] *I think these questions can be improved but I will keep working through the methods and results, then return to this.*

[12] *This is vague. Can you articulate this question more precisely?*

[13] *Could be here or elsewhere, but I recommend including hypotheses or predictions toward the end of the Intro. Readers will need to know what to look for in the results. This occurred to me when I got to the methods section about modeling effects aridity, herbivory, and symbiosis. There is not enough provided here to anticipate what types of analyses you would need to do, and what your expectations would be. So that is the main point of this comment: set up expectations for readers.*

*Epichloë amarillans* (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* can also harbor endophyte taxa capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). Horizontal transmission is rare, influenced by environmental and genetic factors, and was not observed in our study system. To examine the effects of endophytes on host species across a stress gradient, we established eight plots (1.5 m $\times$ 1.5 m) at each of seven sites along an aridity gradient in Louisiana and Texas (Fig. 1A-D). Each plot contained 15 individuals of a single species, with half of the plots protected from herbivores using exclusion cages to assess the interactive effects of endophytes and herbivory (Fig. 1E). In summer–fall 2021, seeds were collected from naturally symbiotic source populations and either heat-treated to remove endophytes or left untreated, producing symbiont-free ($E^-$) and symbiotic ($E^+$) plants from the same genetic lineages.

## Source populations and symbiont removal

Plants used in the common garden experiment were derived from natural (source) populations throughout the natural distribution of the species (Fig. 1). Seeds were collected from these source populations, and some were heat-treated at 60°C for approximately five days (120 hours) to produce endophyte-negative plants ($E^-$), a method that eliminates endophytes without affecting seed viability. All seeds, both heat-treated and non-heat-treated, were planted in the Rice University greenhouse, where seedlings were fertilized every two weeks. Seedlings were then vegetatively propagated to produce enough individuals for the experiment. Before field planting, we confirmed endophyte status ($E^+$ or $E^-$) of all seedlings using either microscopy or an immunoblot assay. Leaf tissues were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and a chromagen to visually indicate presence or absence. Both methods yield similar detection rates.

## Common garden experimental design

We conducted our common experiment across seven sites[14] spanning the precipitation gradient from Louisiana to central Texas, USA (Fig. 1). During our study period (2023-2025) realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites. Mean annual temperature did not differ significantly across

[14] *You should create a table for the lat long of each site and a brief description of the site. This can go in a supplement. Lot/Long is important for meta-analysis, as you know.*

sites (Fig. S-1), so we focus on precipitation as the primary abiotic factor that varied across sites. This gradient also overlaps and exceeds the natural distributions of our focal species, ensuring realistic exposure to climatic and biotic conditions (Fig. 1).

Plots were randomly assigned one of four initial endophyte frequency treatments (80%, 60%, 40%, or 20%) and one of two herbivore access treatments (fenced to exclude herbivores or unfenced to allow access). In the fenced plots, insecticides were applied periodically to eliminate small herbivorous insects, as *Agrostis hyemalis*, *Epichloë amarillans*, and *Poa autumnalis* are naturally subject to herbivory by both insects and larger mammals. To ensure that our herbivory treatment reflected the level of herbivory experienced by individual plants, we counted the number of damaged plants per plot and estimated the proportion of damaged plants per site and herbivore treatment. Our results confirmed that the herbivory treatment was effective (Fig. 1F).

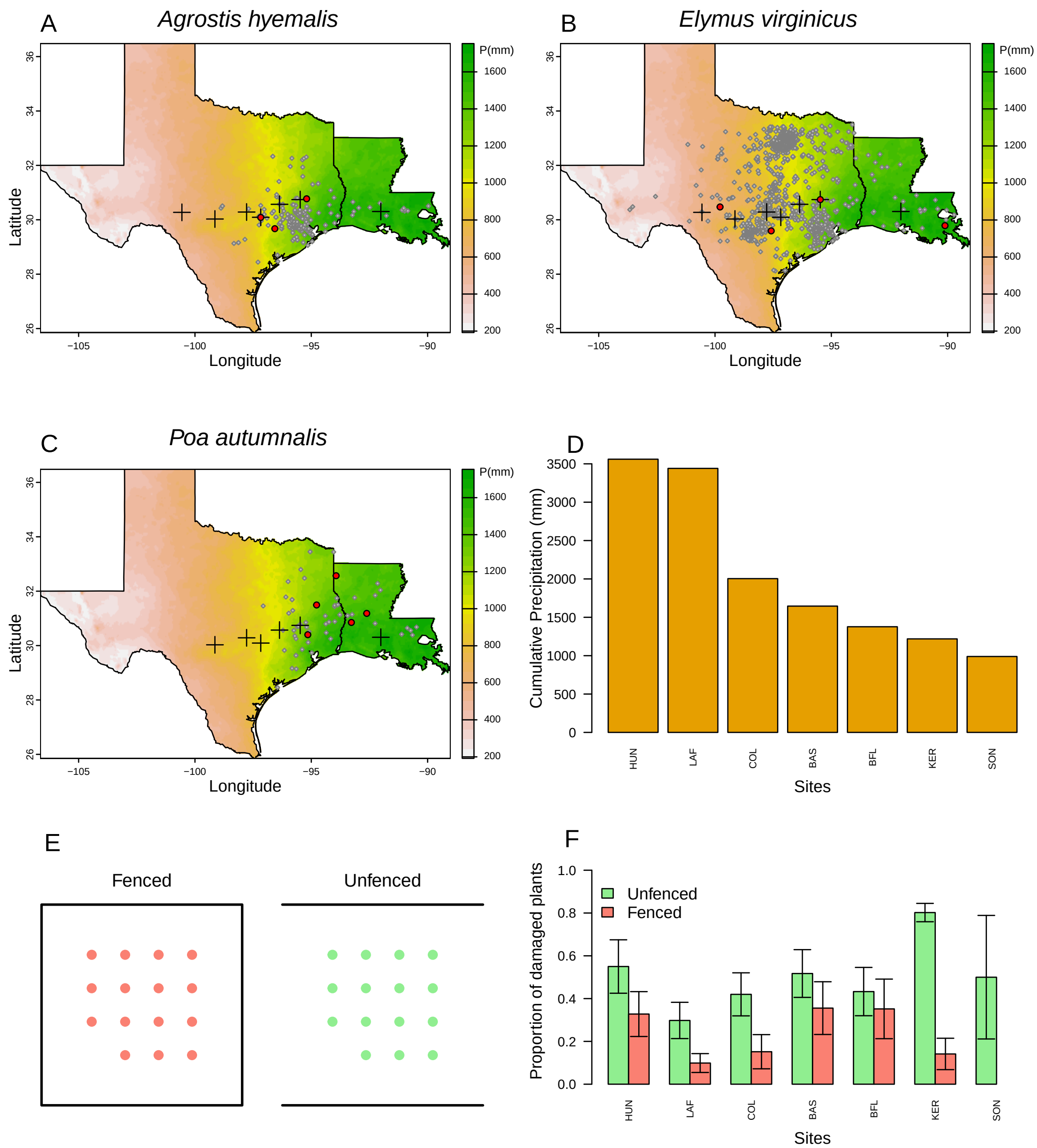


**Figure 1:** Distribution of common garden sites along a natural aridity gradient from Texas to Louisiana, US for (A) *Agrostis hyemalis* (AGHY), (B) *Elymus virginicus* (ELVI), and (C) *Poa autumnalis* (POAU). Climate data in A–C are based on 30-year normals of climate predictions (1993–2023) from PRISM data (PRISM Climate Group, 2024), which illustrate the precipitation gradient across sites (scale on the right is in mm. Red dots indicate the locations of source populations, while grey dots represent Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area (GBIF, 2025a,b,c). D) Cumulative precipitation during the study period (May 2023 – June 2025) at each site, ordered west to east. Values are also derived from PRISM climate data (PRISM Climate Group, 2024).

## Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May, and *Elymus virginicus* was censused in June of 2023, 2024, and 2025. For each individual, plant survival was recorded as alive or dead and, for surviving plants, size was measured as the number of tillers. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*.

## Modeling the effects of aridity, herbivore exclusion, and symbiosis on grass demography

To quantify how abiotic stress and herbivore exclusion (as a proxy for herbivory pressure) modify the demographic outcomes of plant–fungal interactions, we developed hierarchical models for survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, and fertility with the number of inflorescences per plant, and for *E. virginicus* and *P. autumnalis*, the number of spikelets with negative binomial distributions. Growth was quantified as the log ratio of the number of tillers between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in both years. We modeled all vital rates (survival, growth, and fertility) using species-specific fixed effects for symbiosis (endophyte presence), precipitation, and herbivore access, including all relevant two- and three-way interactions. Each observation $i$ corresponds to a plant in a given year; individuals can appear in multiple years. For simplicity, we omit the full subscript notation for species, plot, population, and site-year in the following equation, which is explicitly included in the Stan model:

$$\begin{aligned}\mu_i = {} & \beta_0[Sp_i] + \beta_P[Sp_i]P_i + \beta_{P^2}[Sp_i]P_i^2 + \beta_{endo}[Sp_i]endo_i + \beta_{herb}[Sp_i]herb_i \\ & + \beta_{endo\times P}[Sp_i]endo_i P_i + \beta_{endo\times P^2}[Sp_i]endo_i P_i^2 + \beta_{herb\times P}[Sp_i]herb_i P_i \\ & + \beta_{herb\times endo}[Sp_i]herb_i endo_i + \beta_{endo\times herb\times P}[Sp_i]endo_i herb_i P_i \\ & + \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i],Sp_i} + \rho_{\text{plot}[i],Sp_i}.\end{aligned} \tag{1}$$

Here, $P_i$ is total precipitation (mm) averaged over the 12 months preceding the census, $endo_i$ indicates endophyte presence, and $herb_i$ indicates herbivore access. Random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity not explained by precipitation, $\omega_{\text{source}[i],Sp_i}$ captures variation associated with the provenance of transplants, and $\rho_{\text{plot}[i],Sp_i}$ captures plot-to-plot variation. Each species has its own random effect for each site, source, and plot, while the variance parameters ($\sigma_{\text{site}}$,

$\sigma_{\text{source}}$, $\sigma_{\text{plot}}$) are shared across species. This hierarchical structure allows the model to "borrow strength" across species, improving estimates of fixed effects and background variation, while still capturing species-specific responses to environmental heterogeneity.

## Disentangling abiotic and biotic drivers on cool-season grass demography

To disentangle the relative contributions of abiotic and biotic factors to endophyte–host demography, we developed three candidate models.[15] The first included only abiotic predictors (precipitation), the second included only biotic predictors (herbivory and endophyte status), and the third included both abiotic and biotic predictors. All models followed the same hierarchical structure described above, differing only in the inclusion of predictor terms (Supplementary Method S.1.2). [16]

To determine the best-supported model for each vital rate[17], we compared candidate models using leave-one-out cross-validation (LOOCV) (Vehtari et al., 2017). LOOCV combines validation and training by iteratively holding out one observation as a test set while fitting the model on the remaining $n-1$ observations. This process is repeated $n$ times, once for each observation, resulting in $n$ fitted models (Silva and Zanella, 2024). The overall test error estimate is then obtained by averaging the prediction errors across all $n$ models (Eq. 2).[18]

$$CV_n = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024).

[15] *It is a little confusing which vital rates this model comparison was done for, and if it's all of them, isn't that more than three models?*

[16] *This was a very clever idea and I think it is a great addition to the paper. I think it might make more sense to include endophyte in all the models, because the question is really about whether biotic or abiotic stressors are more important for the benefits of the symbiosis. The models then become (1) endophyte*precip, (2) endophyte*herbivory, (3) endophyte*precip + endophyte*herbivory and/or maybe endophyte*precip*herbivory, which is the three-way interaction I asked about.*

[17] *Not clear from above that this was done for each vital rate. In that case I am not sure what we are looking at in the figure. Is this an average across vital rates? Is that legit?*

[18] *And averaging across vital rates?*

# Results

## Endophyte presence influenced host survival and reproduction across the abiotic stress gradient, with no effects of herbivory

We used Bayesian mixed-effect models to test the effects of abiotic stress (geographic gradient of precipitation), endophyte association ($E^-$ vs. $E^+$), and herbivore exclusion (fenced vs. unfenced plots) on survival, growth, and reproductive traits in three grass host species.[19] These models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles (Fig. S-2). High abiotic stress reduced survival, with $E^-$ individuals outperforming $E^+$ individuals, indicating a cost of endophyte association under dry conditions (Fig. 2).[20] Growth was generally similar across species and treatments, except in *Agrostis hyemalis*, where $E^+$ plants showed higher relative growth along the stress gradient (Fig. **??**). Inflorescence production differed little between $E^+$ and $E^-$ individuals, except for *Poa autumnalis*, which produced more inflorescences under low abiotic stress (high precipitation; Fig. **??**). Spikelet production also reflected costs of endophyte association under high stress, particularly in *Elymus virginicus*, where $E^+$ individuals produced fewer spikelets than $E^-$ individuals (Fig. **??**). Herbivore exclusion had minimal effects on all vital rates, with slightly higher survival and spikelet production in fenced plots, indicating that precipitation and endophyte status were the primary drivers of variation in host demography.[21] [22] [23]

[19] *Rather than provide a recap here, lead with a strong topic sentence that gives an overview of the results described next.*

[20] *I think there are two results here that should be separately acknowledged: survival declined under lower precipitation, and the effect of endophytes on survival changed from beneficial under wetter conditions to costly under drier conditions. I think these results can be communicated more clearly, and you should also provide statistical support for this result.*

[21] *You have not provided any statistical support to justify this. Precipitation generally looks like a stronger effect than endophyte status. But I don't think the relevant comparison is among those three things. I think we are asking whether the effect of endophyte status was more influenced by precipitation, fencing, both, or neither. That is the contrast that I think the text needs to bring into focus.*

[22] *This paragraph packs in a lot of results into very little text. I think the results need more "room to breathe", probably expanded into several paragraphs, maybe one for each vital rate. You do not mention anything about species differences, and I think that is important. For example, in POAU there is basically no endophyte effect on survival, and that result gets completely lost here. You also do not provide any statistical support, nor do you point readers to which specific elements of the figures provide support for the statements you make here. The overall comment here is that this section of the results should be more thorough and more clearly grounded in and supported by statistical results and visualizations.*

[23] *Some comments on the figures. You label columns Unfenced and Fenced, but in the text you describe this as 'herbivory'. I think using the fence language is more accurate and transparent, but I suggest aligning the words you use in the text with the same words in the figures. I am not sure where there are so few data points. We have hundreds of plants, many censused more than once. There should be many more data points. Your figure legends say that the points are means. It could make sense to show means for survival, but for the other vital rates I would show all the raw data. Even for the means, it seems like there should be more points. Also, you say the points indicate plot means, but I don't think this makes sense because plots are composed of both E+ and E-. You should label all panel elements so that you can refer to them in the text, which relates to my comment that there needs to be stronger correspondence between what you say and what you show. Don't put all species on the same y-axis scales, esp in Figure 4. You need communicative y-axis labels on all panels, on the left side of the plot.*

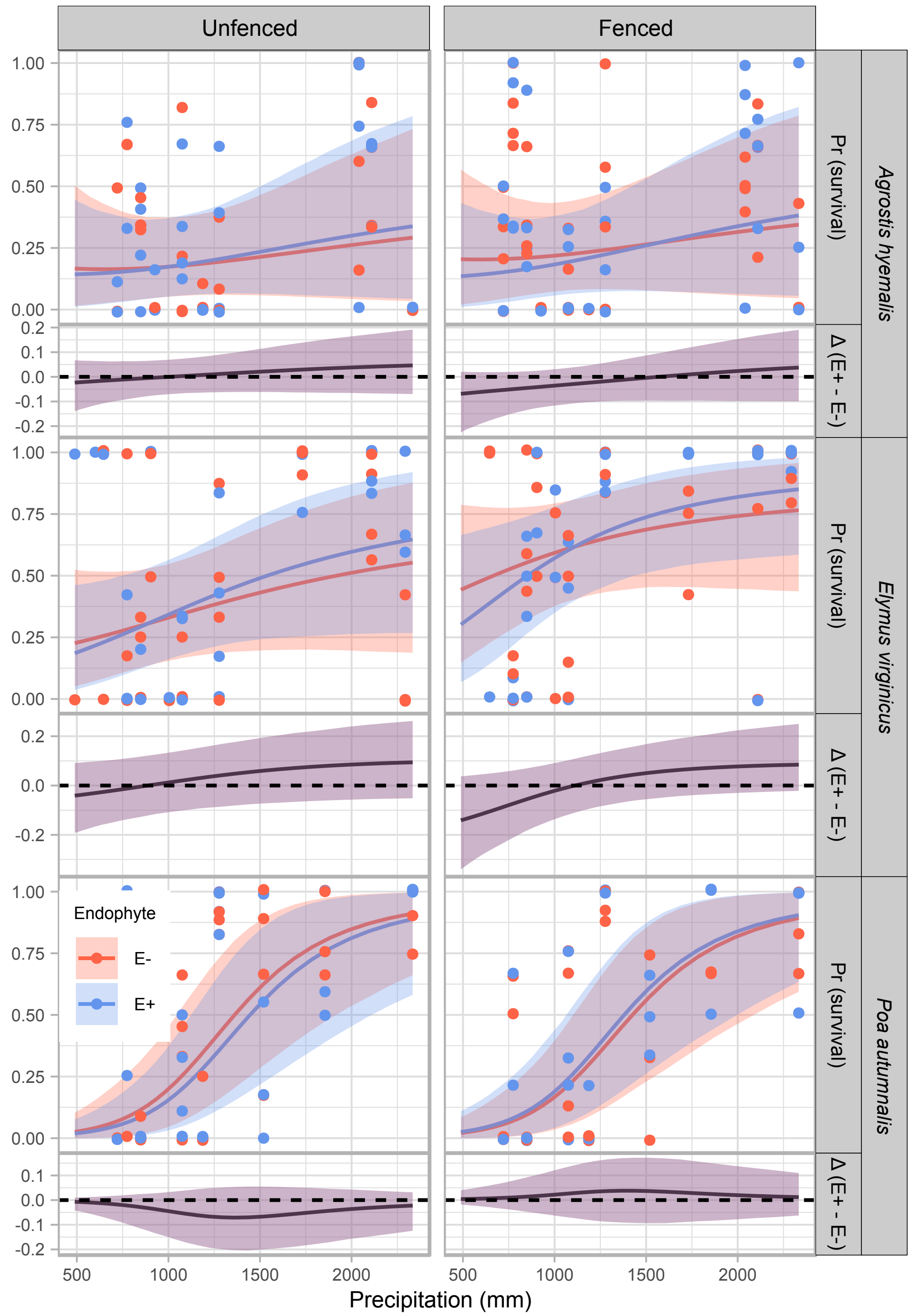


**Figure 2:** Predicted survival rate (top panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for $E^{+}$ (blue) and $E^{-}$ (red), with shading indicating 90% credible intervals. Points represent mean observed survival per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species (AGHY, ELVI, POAU) and herbivory treatment (Fenced, Unfenced).

### Abiotic factors as the primary drivers of fitness components across the stress gradient

Across all vital rates (survival, growth, inflorescence, and spikelet production), models that included only abiotic predictors consistently provided the best fits, measured by expected log predictive density (ELPD; higher values indicate better predictive accuracy; Figure **??**). Uncertainty estimates showed that in some cases the advantage of abiotic-only models over those that also included biotic predictors was small. For example, for survival, the abiotic-only model outperformed the abiotic and biotic model only slightly ($\Delta$ELPD $= -2.1$, SE = 3.7), suggesting that adding biotic predictors contributed little additional explanatory power. By contrast, models based only on biotic factors performed much worse (e.g., survival: $\Delta$ELPD $= -5.2$, SE = 2.0). Taken together, these results suggest that the demography of *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* is shaped primarily by abiotic stress related to precipitation, while biotic interactions (herbivory and endophyte presence) play a secondary role. [24] This pattern aligns with the stress gradient hypothesis, which predicts that abiotic factors dominate in stressful environments, whereas biotic interactions gain importance under more benign conditions.[25] [26]


## Acknowledgements

This research was supported by National Science Foundation Division of Environmental Biology awards.


[24] *Following my comments above, I think this analysis would be better focused if all models included endophyte, and we are comparing endo*precip against endo*herb against endo*precip*herb. I don't think it is particularly meaningful that abiotic stress was stronger than the effect of endophytes – that does not surprise me and in some sense is not the point of the experiment. We really want to know what are the factors that make this symbiosis more mutualistic or more parasitic.*

[25] *I actually don't think it aligns with the SGH for two reasons. First, this analysis is currently saying that precip is more important for demography than biotic interactions. That is not the SGH. The SGH is about whether *species interactions* become more positive under stress. The second reason is that you found the opposite: the symbiosis became *less positive* under stress!*

[26] *I really like this analysis! This is another place where I think readers will need more guidance about what the alternative expectations might be, so that they could interpret this figure appropriately. If these points are averages why does abiotic have no error bars? Is it weird that the abiotic-biotic error bars exceed zero?*

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# Supporting Information

## S.1 Supplementary Methods

[27]

### S.1.1 Study design

#### S.1.1.1 Abiotic stress

Common gardens were established at seven sites spanning the geographic range of *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* from Louisiana to Texas, USA. Sites were selected along a pronounced aridity gradient, with Louisiana representing mesic conditions and Texas increasingly arid conditions. Mean annual temperature did not differ significantly among sites (Fig. S-1), confirming that precipitation is the primary abiotic gradient. Sites were chosen to reflect natural environmental conditions, including shaded areas under tree canopy or near shrubs.

#### S.1.1.2 Plot establishment

At each site, we established eight plots measuring 1.5 m $\times$ 1.5 m. Existing vegetation was removed by tilling to reduce competition. Within each plot, 15 individuals of each species were planted approximately 15 cm deep in an evenly spaced 4 $\times$ 4 grid, with positions randomly assigned to reduce positional effects.

#### S.1.1.3 Endophyte and herbivory treatments

Seeds were collected in summer–fall 2021 from naturally symbiotic populations (source populations). Seeds were either heat-treated to remove endophytes ($E^-$) or left untreated ($E^+$), maintaining the same genetic lineages. Each plot was randomly assigned a starting endophyte frequency (80%, 60%, 40%, or 20%) and a herbivory treatment. For herbivore exclusion plots, plants were enclosed with 1.2 m tall mesh fencing and sprayed with insecticide to prevent vertebrate and insect herbivory. For herbivore access (control) plots, plants were half-enclosed with mesh netting. All plots received comparable quantities of source population material.

[27] *Much of this section repeats verbatim what you have in the main text. I suggest eliminating this section and putting all the field methods in the main document.*

#### S.1.1.4 Experimental rationale

This experimental design was informed by natural patterns of endophyte prevalence in Texas (Donald et al., 2021; Rudgers et al., 2009; Sneck et al., 2017).[28] Fungal genotyping confirmed the presence of biosynthetic pathways for secondary metabolites, such as peramine, loline, and ergot alkaloids, which may enhance host resistance to drought and herbivory (Beaudry, 1951). [29]The design allows testing how endophyte frequency and herbivory interact with abiotic stress (aridity) to influence host demography from core to edge of the species range.

#### S.1.1.5 Source populations and identification of individual endophyte status

Plants used in the common garden experiment were derived from natural populations throughout the south-central US. Seeds were collected from these populations, and some were heat-treated at 60°C for approximately five days (120 hours) to produce endophyte-negative plants ($E^-$), a method that eliminates endophytes without affecting seed viability. All seeds, both heat-treated and non-heat-treated, were planted in the Rice University greenhouse, where seedlings were fertilized every two weeks. Seedlings were then vegetatively propagated to produce enough individuals for the experiment (N = 840). Before field planting, we confirmed endophyte status ($E^+$ or $E^-$) of all seedlings using either microscopy or an immunoblot assay. Leaf tissues were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and a chromagen to visually indicate presence or absence. Both methods yield similar detection rates.

### S.1.2 Candidate models for abiotic and biotic drivers of cool-grass species

To disentangle the relative contributions of abiotic and biotic factors, we developed three candidate models:

1. **Abiotic model:** includes precipitation ($P$) and its quadratic term ($P^2$) as predictors. The expected value of each vital rate for individual $i$ is:

$$\mu_i = \beta_0 + \beta_P P_i + \beta_{P^2} P_i^2$$
$$+ \phi_{\text{site}[i]} + \omega_{\text{source}[i]} + \rho_{\text{plot}[i]}$$

[28] *Informed how? I am not sure about the purpose of this section.*

[29] *Unclear what genotyping you are referring to. I don't know the paper cited here but they certainly did not genotype anything in 1951.*

2. **Biotic model:** includes endophyte status (endo), herbivory (herb), and their interaction (endo $\times$ herb):

$$\mu_i = \beta_0 + \beta_{\text{endo}} endo_i + \beta_{\text{herb}} herb_i + \beta_{\text{endo}\times\text{herb}} endo_i herb_i$$
$$+ \phi_{\text{site}[i]} + \omega_{\text{source}[i]} + \rho_{\text{plot}[i]}$$

3. **Full model:** combines both abiotic and biotic predictors, including all relevant interactions:

$$\mu_i = \beta_0 + \beta_P P_i + \beta_{P^2} P_i^2 + \beta_{\text{endo}} endo_i + \beta_{\text{herb}} herb_i$$
$$+ \beta_{\text{endo}\times P} endo_i P_i + \beta_{\text{herb}\times P} herb_i P_i + \beta_{\text{herb}\times\text{endo}} herb_i endo_i + \beta_{\text{endo}\times P^2} endo_i P_i^2$$
$$+ \phi_{\text{site}[i]} + \omega_{\text{source}[i]} + \rho_{\text{plot}[i]}$$

All models included a hierarchical random effects structure accounting for site, source, and plot variation, with $\phi_{\text{site}[i]} \sim \mathcal{N}(0, \sigma_{\text{site}})$, $\omega_{\text{source}[i]} \sim \mathcal{N}(0, \sigma_{\text{source}})$, and $\rho_{\text{plot}[i]} \sim \mathcal{N}(0, \sigma_{\text{plot}})$.

## S.2 Supporting Tables

**Table S-1:** Populations and Coordinates

| Population | Location | Coordinates | NE+ / NE- |
|---|---|---|---|
| Sam Houston State University Field Station | Huntsville, TX | 30°44′30.5″N, 95°28′28.2″W | NE+ = 39, NE- = 50 |
| Texas Tech University | Junction, TX | 30°28′18.2″N, 99°47′01.7″W | NE+ = 49, NE- = 83 |
| Palmetto State Park | Palmetto, TX | 29.591623, -97.584781 | NE+ = 250, NE- = 192 |
| Jean Lafitte National Historical Park and Preserve | Jean Lafitte, LA | 29.785105, -90.114933 | NE+ = 83, NE- = 94 |

*I need to clean this Table and add the remaining sites* *Different species came from different source populations, so that should be clarified.*

**Table S-2:** Differences in expected log predictive density (ELPD) between models for each vital rate. The top model for each response (bold) has the highest ELPD. Standard errors of the differences are shown in the `se_diff` column.

| **Vital rate** | **Model** | **elpd_diff** | **se_diff** |
|---|---|---:|---:|
| Survival | **abiotic** | **0.0** | **0.0** |
| | abiotic_biotic | -2.1 | 3.7 |
| | biotic | -5.2 | 2.0 |
| Growth | **abiotic** | **0.0** | **0.0** |
| | abiotic_biotic | -2.1 | 3.7 |
| | biotic | -5.2 | 2.0 |
| Inflorescence | **abiotic** | **0.0** | **0.0** |
| | abiotic_biotic | -2.1 | 3.7 |
| | biotic | -5.2 | 2.0 |
| Spikelet | **abiotic** | **0.0** | **0.0** |
| | abiotic_biotic | -2.1 | 3.7 |
| | biotic | -5.2 | 2.0 |

# S.3 Supporting Figures

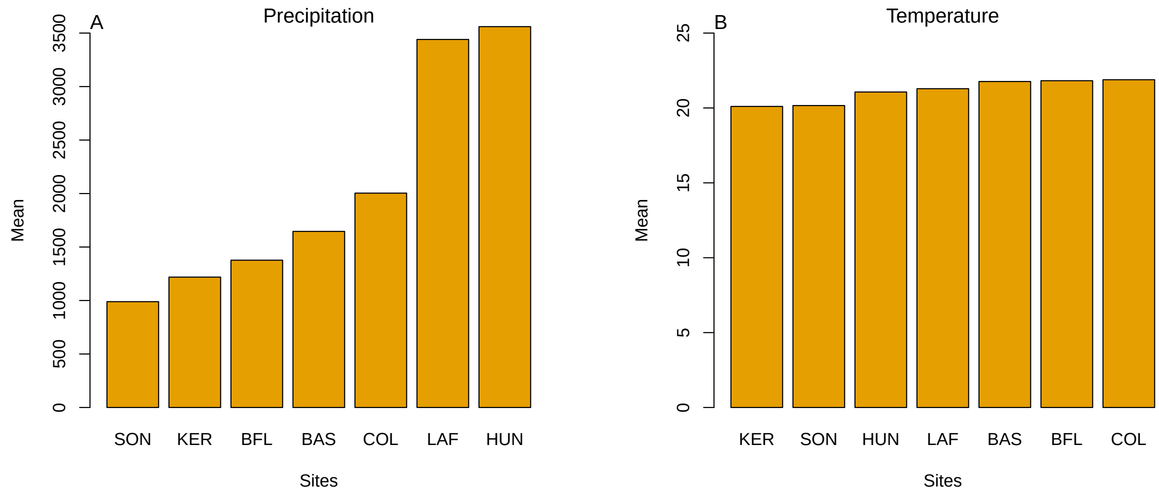


**Figure S-1:** Abiotic conditions across the seven study sites. (A) Mean annual precipitation showing the pronounced east–west aridity gradient from mesic Louisiana to arid Texas. (B) Mean annual temperature across sites, which does not differ much. Panels highlight that precipitation, rather than temperature, constitutes the primary abiotic gradient in this system.

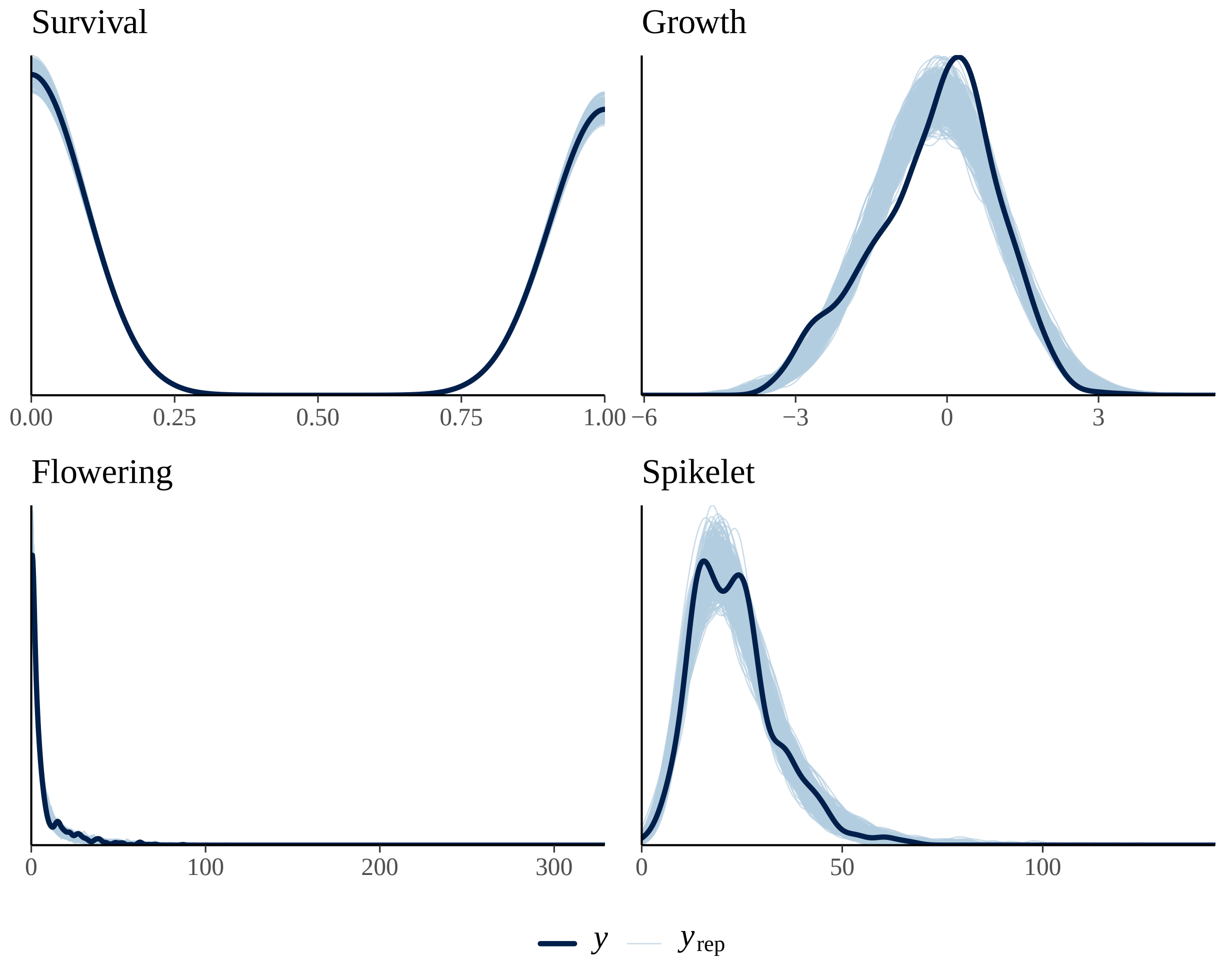


**Figure S-2:** Posterior predictive checks for the full demographic model. Dark blue lines show observed data ($y$), and light blue lines show replicated datasets generated from posterior predictions ($y_{\text{rep}}$). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.